

Thirayu Meererksom

Technology Leader · AI Researcher · Digital Transformation Consultant

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PROFESSIONAL SUMMARY

Ph.D. researcher and full-stack technology leader with 15+ years bridging advanced technology (AI & software architecture) and business strategy (MBA). Proven record delivering national-scale AI healthcare platforms for Thai government agencies and enterprise solutions for global clients in Switzerland and the USA. Peer-reviewed author published in Scientific Reports (Nature portfolio).

EDUCATION

Ph.D. Candidate, Translational Medicine (International Program)

2018–Present

[Suranaree University of Technology](#)

Dissertation: Detection and Characterization of Focal Liver Lesions on CT Images Using Deep Learning. GPAX 3.72

Master of Business Administration (MBA)

2008–2010

[Khon Kaen University](#)

IS: Strategies for Improving the Production Cost of a Software House.

B.Sc. Computer Science

2004–2008

[Khon Kaen University](#)

Project: A Patient Navigation System in Hospital using RFID.

PROFESSIONAL EXPERIENCE

Biz & Tech Consultant

Aug 2019–Present

[KASIRA HOUSE Co., Ltd.](#)

- Directed national-scale AI healthcare platforms (AI-OVCCA, AI-Malaria, AI-Snaily) across provinces with NECTEC & the Ministry of Public Health.
- Designed scalable microservices (Docker, Kubernetes, Kong) and led KH-RMS hospital & KH-School delivery on .NET Core 8 + Angular.
- CyberSec Consultant at Big Boom Asia Co., Ltd.

Researcher

Aug 2019–Present

[Parasitic Disease Research Center, SUT](#)

- Researcher at PDRC (SUT)

Senior Technical Partner (Remote)

2009–Present

[PEZZ AG, Switzerland](#)

- Sole technical partner for 15+ years; modernized legacy systems to cloud SaaS and iOS/Android with ~99.9% uptime across Europe.

Lecturer & Special Project Lead

2011–2019

[Nakhon Ratchasima Rajabhat University](#)

- Owned the B.B.A. Business Computer curriculum; led university digital projects (e-Learning, MIS) and mentored developers.

Academic Staff (Educational Technologist)

2010–2011

[Rajamangala University of Technology Isan](#)

Project Manager

2009

[Cimatt Business Group Co., Ltd.](#)

.NET Senior Developer

2008–2009

[Miracle Inspire Co., Ltd.](#)

.NET Developer

2008

[Genius Megasofts Co., Ltd.](#)

CONSULTING & ADVISORY

Registered Thai Consultant

2024–Present

SME Coach (in-depth consultant) 2022–Present

Thai SME-GP [ISMED]

Marketing strategy, government procurement, ISO, finance/funding. · Clients: Bioway, Dairy Home, K.P. Generation, S-Camp, Coffee Roasting, etc.

Speaker & in-depth consultant 2023–Present

Thai SME-GP [SSRN: Isan + Northern]

Gov procurement, MarTech, SME ONE ID, Google Business, Canva, photography. · ภาคอีสานและภาคเหนือ

Speaker & in-depth consultant 2024–Present

Thai SME-GP [SSRU Isan]

Gov procurement, MarTech, ISO/GMP/MiT, Google Business, Canva + continued MSME consulting. · ที่ปรึกษาต่อเนื่องแก่ MSME 10+ ราย

Freelance Business & Digital Consultant 2022–2024

Independent

Online marketing, import–export, packaging design. · Clients: Coffee Roasting, Chamoke

Business Consultant 2019–Present

KASIRA HOUSE Co., Ltd.

Digital transformation & business strategy.

Cybersecurity Consultant 2019–Present

Big Boom Asia Co., Ltd.

Information security advisory.

Expert Lab Verification (Helminths) 2020–Present

Department of Disease Control (กองโรคติดต่อทั่วไป), MoPH

Stool helminth verification by Formalin-Ether Concentration Technique · via Kasira House

INVITED SPEAKING & TRAINING

SMART TEACHER SMART KIDS WITH AI (4 regions) May–Jun 2026

Chantawapi Co. / Thai Media Fund — Media & AI literacy for early-childhood teachers

Student Readiness Program, Faculty of Public Health 10 Sep 2025

Vongchavalitkul University — Smart Public Health & Occupational Safety: Using Technology & AI Wisely

7th National Local Scout Jamboree — Life Skills: E-Sport 6–10 Aug 2025

Khonburi School, Nakhon Ratchasima PAO — E-Sport skills training

Career-readiness workshop (Digital Media program) 1 Apr 2025

Nakhon Ratchasima Rajabhat University — Job-hunting techniques & essential skills in the digital era

True Community Builder 2024 - 2025 Jan 2024–Dec 2025

TRUE Digital — True Digital Community Builder 2024 - 2025

SME Restart Oct 2023

Suan Sunandha Rajabhat University — Green Tourism for community-based tourism

AI for MSME Oct 2022–Sep 2025

Office of SMEs Promotion (OSMEP)

BDS Market Nov 2022–Jan 2026

Suan Sunandha Rajabhat University — SME Business Development Services (BDS) Market

Thai SME-GP Jan 2022–Jan 2026

Suan Sunandha Rajabhat University

ETDA DIGITAL CITIZEN (EDC) Jan 2022–Present

ETDA — ETDA Digital Citizen (EDC)

Artificial Intelligence for Automated Microscopic Helminth Screening Oct 2020–Present

Department of Disease Control

Workshop — Long Term Care (3C) & Blockchain health records 17 Jan 2020

OTOP Nawatwithi Community-Based Tourism (Nakhon Ratchasima and Buriram)

Jan 2019–Present

Ministry of Social Development & Human Security

AI in healthcare

Oct 2018–Sep 2024

Ministry of Public Health

Government & partner organizations served

Office of SMEs Promotion (OSMEP), Ministry of Public Health, Ministry of Higher Education, Science, Research & Innovation (MHESI), Ministry of Social Development & Human Security, Department of Disease Control, Department of Health, Thai Media Fund, ETDA, TRUE Digital, NECTEC, Ministry of Commerce (OTOP projects), Ministry of Interior — CDD (OTOP Village)

Selected training topics

- AI & Computer Vision (helminths, malaria, leprosy, snail detection)
- Deep Learning & NLP (e.g., human-trafficking surveillance)
- Digital & social-media marketing technology
- Line MyShop, Shopee, Lazada, TikTok Shop
- Mobile & web application development
- CMS: Joomla, Mambo, Moodle, WordPress
- OTOP product development & online marketing
- E-commerce for all (E&M Commerce)
- Business Model Canvas & related tools
- Content creation for content marketing
- Metaverse for education technology
- Smart tourism & geopark information (Khorat Geopark)
- MVC concept; UTP & basic networking
- Fleet management system for truck companies
- Rice co-operative logistics management
- ITPE knowledge for advanced IT
- Thailand World Skill — Micro-Computer Level 1
- E-Sport skills & competition
- Thai SME-GP and BDS Market

Documented engagements

- 10 Sep 2025 — Vongchavalitkul University (Public Health) — Speaker — Smart Public Health & Occupational Safety with AI
- 6–10 Aug 2025 — Khonburi School / Korat PAO — Trainer — E-Sport (Korat Esports VP)
- 1 Apr 2025 — Nakhon Ratchasima Rajabhat University (Digital Media) — Speaker — job-hunting & digital-era skills
- 31 Jan 2020 — Department of Disease Control (กองโรคติดต่อทั่วไป) — Helminth stool verification (Formalin-Ether) — Kasira House
- 17 Jan 2020 — Department of Health (กรมอนามัย, สำนักอนามัยผู้สูงอายุ) — Speaker — Long Term Care (3C)/Care Manager curriculum, Chiang Mai
- - — School board committee — Invitation — school board (Children's Day)

SELECTED PROJECTS & SYSTEMS

AI & Healthcare

AI-Snaily — Real-time YOLO detection of liver-fluke host snails (98.7% accuracy) — basis of the 2026 Scientific Reports paper.

AI-OVMIF / AI-Helminths — Parasite-egg detection AI (OV, MIF, Cs, Ec, AI, Tt, Tsp) for DDC and NECTEC.

AI Malaria & AI Leprosy — Computer-vision diagnosis tools for DDC and Rajpracha Samasai Institute.

Anamai Chatbot & COVID-19 System — Rapid-response healthcare chatbot and pandemic info platform for the Dept. of Health.

SUT OVCCA / Kunióm — Liver-fluke/CCA and elderly-fall self-risk screening tools.

3C & Long-Term Care — Web portal and GIS reporting system for the Dept. of Health.

Enterprise & FinTech

KH-RMS & KH-School — Hospital and school management platforms.

Secure Financial CRM & Dashboard — Mortgage/auto/credit tracking with bank-grade CI/CD (NDA).

ERP & Logistics + POS — Laravel + Angular ERP with SARIMA demand forecasting.

Algorithmic Trading & Automation — MT5/crypto trading bots and n8n/Make automation.

Global / PEZZ

GAS2008 / GAS2010 — Growth Analysis System for PEZZ AG, Switzerland.

Child-Growth / Growth-Guide / Bedwetting / PEZZPat — iOS/Android clinical apps for PEZZ AG.

Mobile & Web

Clear Mechanic (USA) — Android repair-order app over RESTful web services.

Khorat Geopark / Go Khorat — Cross-platform smart-tourism information for the UNESCO Global Geopark.

MegaDict / Thai Talk 2.0 / Pro Balancer — Desktop applications (C#, Speech API).

Enterprise & FinTech

KH-MICE — Korat MICE city platform (web & mobile)

KH-Form — Form builder beyond Google Forms, with Passkey auth
Korat Esport Match — Esports tournament data & challenge platform
Game Dashboard — JavaScript game analytics dashboard
Odoo API Integration — Odoo-WinSpeed integration via FastAPI

AI & Healthcare

AI-Teania — Parasite-egg (Tspp) detection AI
Malaria Online — Malaria exam & e-certificate platform
MG-OVCCA — OV-CCA risk-screening questionnaire (LINE LIFF)

Research & Systems

SUT-RDMS — Research Project Management System
SUT-IMPM — Laboratory & research-unit management system
ISAN Goat — Goat-farm research questionnaire (LINE LIFF)

AI & Healthcare

License Plate Recognition — AI license-plate recognition on edge devices

Community

TPA Career Path — Thai Programmer Association career-path roadmap

Enterprise & FinTech

ERP (PPCeramic) — POS & inventory ERP

Storm Manager — Developed a cost management and human resource management system that occurred in the USA, or Storm Manager in the form of a Web Application and Web Service, which includes the following features: • Event Recording System • Workforce Planning and Cost System • Map System • Accommodation Management System • Material and Equipment Management System • Financial System • Summary Report

ERP (Connect Diagnostics) — ERPNext Implementation for Medical Devices Company

ERP (World Fert) — ERP for Fertilize Wholesaler

RESEARCH PUBLICATIONS

ORCID 0000-0001-6414-3607

International Peer-Reviewed Journals

1. Jenwithee T, Meerersom T, Limpanont Y, Sripa B, Laha T, Suwannatrai AT. (2026). Automated real-time surveillance of Bithynia snails using a comparative YOLO based approach for liver fluke host detection. Scientific Reports, 16, 14886. doi.org/10.1038/s41598-026-43387-x
2. Kaewpitoon N, et al. (incl. Meerersom T). (2026). Advancing Diagnostic Accuracy in Liver Cancer: A Systematic Review of Artificial Intelligence Applications. Asian Pacific Journal of Cancer Prevention, 27(1), 5–17. doi.org/10.31557/APJCP.2026.27.1.5
3. Meerersom T et al. (2026). Balancing innovation and interpretation: evaluating artificial intelligence in adenoma detection. Wiener Klinische Wochenschrift. doi.org/10.1007/s00508-025-02599-3
4. Meerersom T et al. (2025). Beyond Accuracy: Advancing ML-Guided Diagnosis of Obstructive Jaundice Through Clinical Integration. Clinical and Translational Gastroenterology. doi.org/10.14309/ctg.0000000000000920
5. Meerersom T et al. (2025). Toward Clinical Readiness: Critical Reflections on PATHOMIQ_PRAD and Artificial Intelligence Histology. JCO Clinical Cancer Informatics. doi.org/10.1200/CCI-25-00227
6. Meerersom T et al. (2025). Rethinking Machine-Learning Metrics in Schistosomiasis Control: Toward Predictive Equity, Field Readiness, and Sustainable Deployment. American Journal of Tropical Medicine and Hygiene. doi.org/10.4269/ajtmh.25-0470a
7. Bukkhunthod P, Meerersom T, et al. (2020). Animation as Supplementary Learning Material About Carcinogenic Liver Fluke in Classes for Primary Schoolchildren. Journal of Cancer Education, 35(1), 14–21. doi.org/10.1007/s13187-018-1434-5
8. Kaewpitoon SJ, Ponphimai S, et al. (incl. Meerersom T). (2019). The prevalence of intestinal helminth infection in rural subdistricts of northeastern Thailand. Tropical Biomedicine, 36(1), 152–164.
9. Kaewpitoon N, et al. (incl. Meerersom T). (2018). Detection of the carcinogenic liver fluke risk groups by verbal screening questionnaires using a mobile application. Asian Pacific Journal of Cancer Prevention. doi.org/10.22034/APJCP.2018.19.7.2013
10. Kaewpitoon SJ, et al. (incl. Meerersom T). (2018). A carcinogenic human liver fluke infection among migrant workers in NE Thailand. Tropical Biomedicine, 35(2), 453–463.
11. Kaewpitoon N, Kaewpitoon SJ, Meerersom T, et al. (2018). Detection of Opisthorchis viverrini Infection among the ASEAN Population in Thailand Using a Verbal Screening Test and Fecal Concentrator Kit. Iranian Journal of Parasitology, 13(2), 258–266.
12. Kaewpitoon SJ, Sangwalee W, et al. (incl. Meerersom T). (2018). Active screening of gastrointestinal helminth infection in migrant workers in Thailand. Journal of International Medical Research. doi.org/10.1177/0300060518786911

13. Kaewpitoon N, Meerersom T, et al. (2021). OVCCA Web Application as Supplementary Material to Facilitate Health Literacy Regarding Carcinogenic Liver Fluke. Asian Pacific Journal of Cancer Prevention, 22(9), 3045–3050. doi.org/10.31557/APJCP.2021.22.9.3045

Conference Proceedings

1. Meerersom T, Chan-aran S, Kaewpitoon SJ, Rujirakul R, Kaewpitoon N. (2016). SUT OVCCA Application: a high-potential tool for screening liver fluke and cholangiocarcinoma. 3rd Asian Symposium on Healthcare Without Borders, Fukuoka, Japan, pp. 90–96.
2. Meerersom T. (2017). Health Application Development in Real-Life. 4th Asian Symposium on Healthcare Without Borders, Hiroshima, Japan, pp. 23–32.
3. Meerersom T, Siripornmongkon W. (2018). Kunlóm: A self-risk assessment for falling among the elderly. 4th Conference on Public Health in Asia, Hiroshima, Japan.
4. Kaewpitoon N, Meerersom T, et al. (2018). Effectiveness of the SUT-OVCCA mobile application for risk-group screening among rural populations. 5th International Congress on Infectious Diseases, Berlin, Germany.
5. Kaewpitoon SJ, ..., Meerersom T, et al. (2018). Prevention and control of carcinogenic liver fluke in rural communities through an inter-/trans-disciplinary university. 5th International Congress on Infectious Diseases, Berlin, Germany.
6. Kaewpitoon SJ, ..., Meerersom T. (2017). LaHanNa Project. 4th Engagement Thailand Annual Conference, Chulalongkorn University, Bangkok.
7. Kaewpitoon N, Meerersom T, et al. (2017). A simple screening for liver-fluke risk group through a smartphone application. Annual Conference, Institute of Medicine, SUT.
8. Meerersom T, Kaewpitoon SJ, Kaewpitoon N. (2016). SUT OVCCA iOS Application. National Conference, Institute of Medicine, SUT.

National Publications (Thai)

1. ธีรยุ มฤกษ์สม. (2555). การปฏิรูปการเรียนการสอนของคณะวิทยาการจัดการ มรภ.นครราชสีมา โดยใช้การจัดการความรู้: กรณีศึกษารายวิชาโครงสร้างข้อมูล.
2. ธีรยุ มฤกษ์สม. (2556). การศึกษาผลการจัดการเรียนรู้ด้วยเทคนิคการสอนแนะ รายวิชาสารสนเทศด้านการเงินและการธนาคาร.
3. ธีรยุ มฤกษ์สม. (2557). การพัฒนาผลสัมฤทธิ์ทางการเรียน วิชาระบบสารสนเทศเพื่องานธุรกิจ ด้วยทฤษฎีการสร้างองค์ความรู้.
4. ธีรยุ มฤกษ์สม. (2558). การจัดการเรียนรู้ด้วยปัญหาเป็นฐาน รายวิชาความมั่นคงของระบบสารสนเทศ.
5. ธีรยุ มฤกษ์สม. (2558). การประเมินความพึงพอใจของประชาชนต่อ อบต.หนองค่าย อ.ประเทย จ.นครราชสีมา.
6. ธีรยุ มฤกษ์สม. (2559). การประเมินความพึงพอใจของประชาชนต่อ อบต.วังไม้แดง อ.ประเทย จ.นครราชสีมา.

TECHNICAL SKILLS

Frontend: HTML5, CSS3/SASS, JavaScript, TypeScript; Angular, React/Next.js, Vue/Nuxt.js; Webpack/Vite; Bootstrap, Tailwind, Material UI

Backend: .NET / .NET Core (C#, VB.NET), PHP (Laravel), Python (FastAPI/Flask), Node.js/Express, Java; REST & WCF Web Services, LINE LIFF/Messaging API

AI / Data: Computer Vision (YOLO, TensorFlow/PyTorch), Deep Learning, NLP, Data Science, SARIMA/Time-Series forecasting, Business Intelligence

Databases: MS SQL Server, MySQL, PostgreSQL, Oracle; MongoDB, CouchDB, Elasticsearch; Neo4j (graph); RabbitMQ; Redis

Cloud / DevOps: Docker, Kubernetes, Kong API Gateway, NGINX, Portainer; Azure, AWS, GCP; CI/CD (Azure DevOps, GitHub Actions); VMware; Linux / Windows Server / macOS

Mobile: Android (Java), iOS (Objective-C, Swift); Flutter/Dart, React Native, Ionic; Unity

Business / Other: Agile & Waterfall, BA/SA, System & Platform Architecture, UI/UX, ERP (Odoo, ERPNext), IoT/Rfid, MarTech, Adobe Creative Cloud, SME Coaching

CERTIFICATIONS & LICENSES

- Super AI Engineer, Level 1 (AIAT) [22p21i0089]
- Microsoft Certified Professional (MCP) & Application Developer (MCAD) [3622775]
- Sun Certified Java Programmer (SCJP), Java 2 Platform 1.4
- Information Technology Passport Examination (ITPE) [THIP15A0161]
- National Skill Standard — Micro-Computer Technician, Level 1 (Ministry of Labour) [บพร.1-4-21-001-0013-57]
- ThaiSP: Service Provider [SP57X95]
- Certified Thai SME Coach [8307]
- Public Debt Management Office (PDMO), Ministry of Finance — Thai Consultant Database [หมายเลข 6481 ระดับ 3]

PROFESSIONAL MEMBERSHIPS & LEADERSHIP

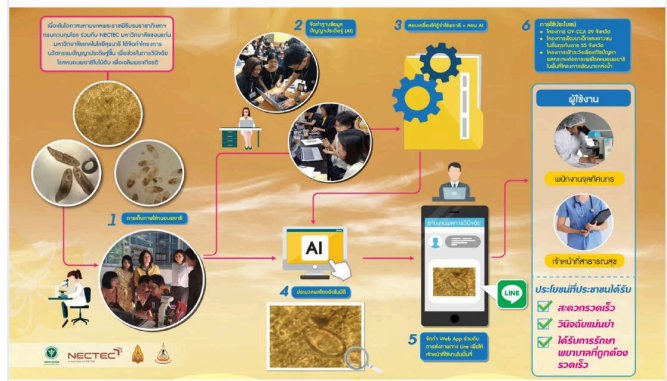
- **Vice President** — Korat Esports Sports Club, Nakhon Ratchasima
- **Committee Member** — Thai Programmer Association (TPA)
- **Member** — Artificial Intelligence Association of Thailand (AIAT)

- **Researcher** — Parasitic Disease Research Center, SUT

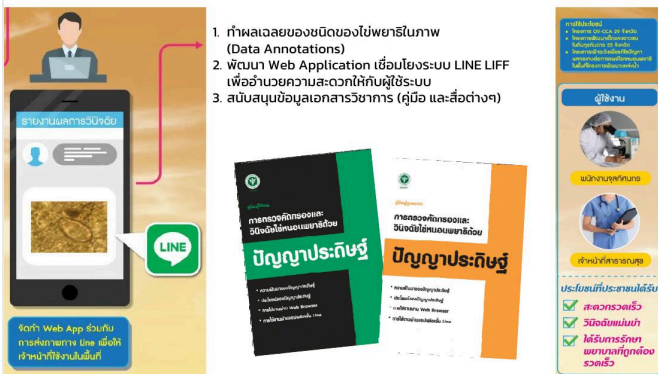
SELECTED WORK — PORTFOLIO



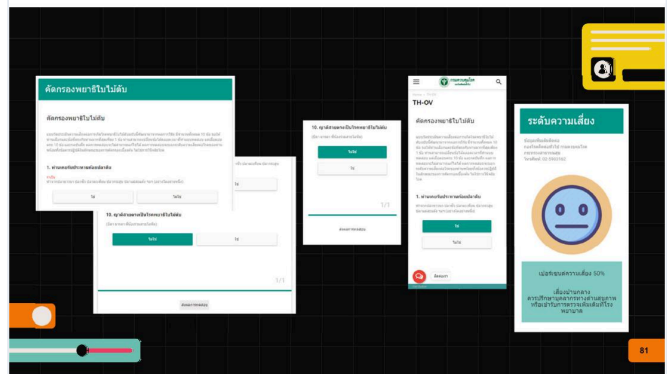
SUT-OVCCA: Liver-fluke/CCA self-risk screening tools and Kunióm: elderly-fall self-risk screening tools.



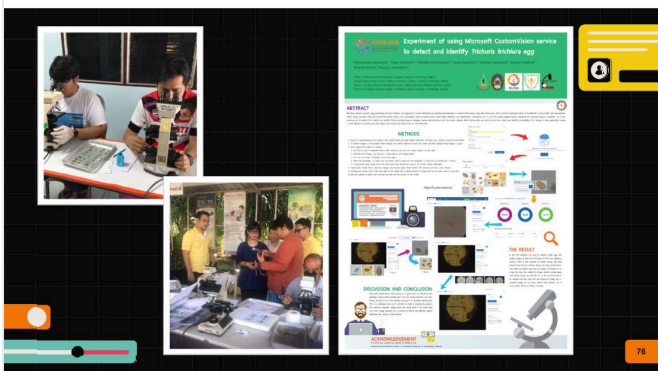
AI-Helminths: Parasite-egg detection AI (OV, MIF, Cs, Ec, AI, Tt, Tspp) for DDC and NECTEC.



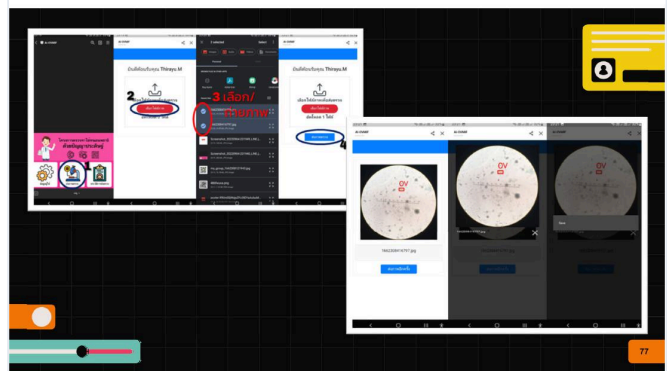
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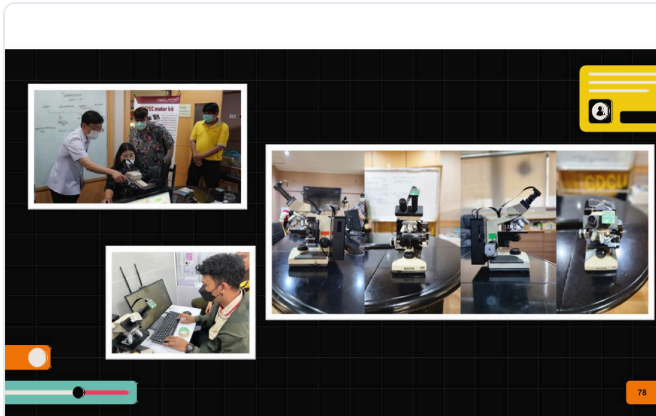
MG-OV-CCA: OV-CCA risk-screening questionnaire (LINE LIFF)



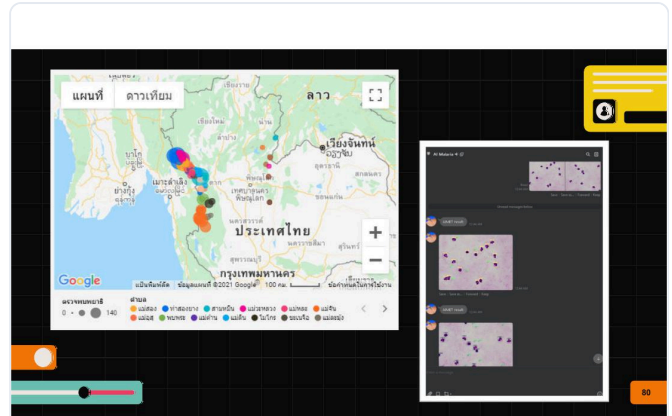
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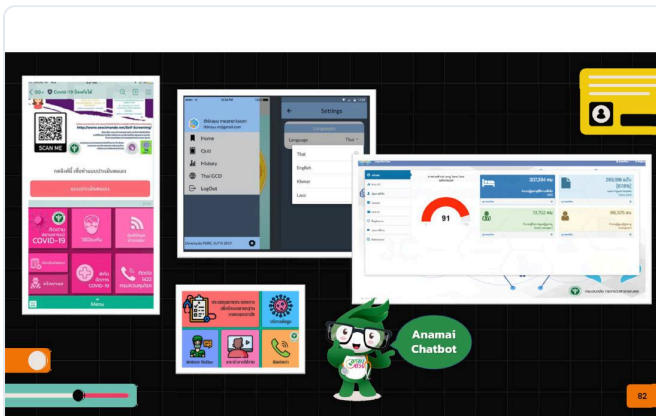
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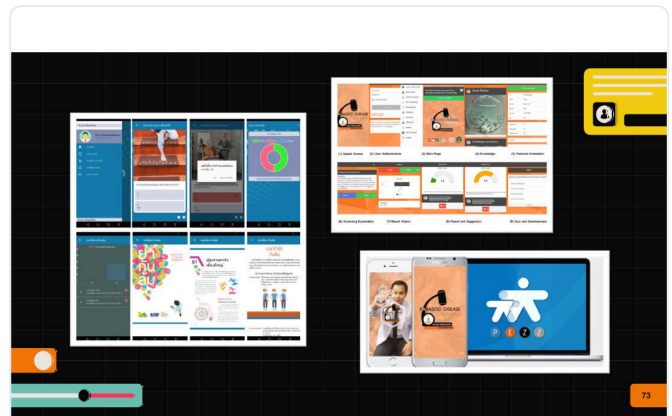
AI-Helminths: Parasite-egg detection AI (OV, MIF, Cs, Ec, Al, Tt, Tspp) for DDC and NECTEC.



AI Malaria: Computer-vision diagnosis tools for DDC. AI-Helminths: Parasite-egg detection AI (OV, MIF, Cs, Ec, Al, Tt, Tspp) for DDC and NECTEC.



3C & Long-Term Care: Web portal and GIS reporting system for the Dept. of Health. Anamai Chatbot: Rapid-response healthcare chatbot for the Dept. of Health.



PEZZ growth-analysis system & clinical apps (GAS, Bedwetting)